Discussion of *The Disclosure Channel of Knowledge Spillovers: Evidence from Environmental Innovation*

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This paper: What it does

- Studies the channels of knowledge diffusion following innovation activity
- **Starting point:** the spillover literature documents that firms benefit from peers' R&D investment, but ignores firms' *disclosure* decisions (patent vs. trade secret)
- Augments Bloom, Schankerman, and Van Reenen (2013) with quality-weighted clean patent spillovers alongside R&D-based measures
- Instruments disclosure using EPA enforcement actions against peer firms, which induce clean patent filings without changing underlying R&D investment

This paper: What it finds

- Knowledge spillover gains dominate business-stealing losses
- Effect driven by process-based clean patents (product-based clean patents and other patents show no incremental spillovers beyond R&D investment)
- **Lower bound:** total social benefits exceed the market-value margin (private returns)

This paper: Why is it important?

The patent system plays a **dual role** in the innovation literature:

- **Incentive role** (Romer, Aghion-Howitt): temporary monopoly rent justifies *ex ante* costly and risky R&D investment
- **Dissemination role** (Mokyr): in exchange, the inventor publishes the invention and knowledge diffuses to the scientific and industrial community

$\longrightarrow$ BSV bundles both roles together, this paper **disentangles** them

$\longrightarrow$ Recent empirical literature not only focuses on innovation, but also on the diffusion of knowledge and technologies, through firms' networks (Bilgin, Faia & Ottaviano, 2025) or interpersonal networks for AI diffusion (Aghion, Bunel, Jaravel, Mikaelsen, Roulet & Sogaard, 2026)

General Comment

A great paper with two important and complementary contributions:

- **Disentangles two margins of the patent system:** the incentive to innovate (R&D investment) and the decision to make knowledge publicly available (disclosure)
- **Uncovers concrete underlying mechanisms** in the context of green innovation

Comment #1: Contribution

The paper departs from BSV along **two distinct dimensions simultaneously**: (i) R&D investment vs. patents; (ii) clean vs. non-clean

Suggestion: two sequential steps would be clearer

- **Step 1 - BSV + patents:** do patents generate incremental spillovers beyond R&D?
- **Step 2 - clean vs. non-clean:** is the channel specific to clean patents? And why?

The current presentation conflates both steps, making it harder to assess which dimension is driving the results

Comment #2: R&D stock and patents

R&D stock (input) and clean patents (output) enter the same regression:

- **Knowledge produced**, which may remain secret (captured by R&D stock)
- **Share of knowledge publicly disclosed**, which generates spillovers (captured by patents)

Concern: R&D stock is not a perfect proxy for knowledge produced under **heterogeneous returns**

- Same R&D investment $\not\Rightarrow$ same knowledge produced (frontier vs. incremental, high- vs. low-risk)
- High-risk firms more likely to patent a successful innovation $\rightarrow$ patent coefficient mixes **innovation quality** and **disclosure**

Partial answer: IV (EPA enforcement) holds R&D constant at $t = 0$ (Table B14)

Comment #3: Asymmetry between input and output

The paper distinguishes clean from non-clean patents (output) but *cannot* make the same distinction for R&D (input)

Controlling for total R&D, residual variation in clean patents reflects *three* things:

- The **disclosure decision**: patent vs. trade secret ✓ identified
- The **composition of R&D**: share allocated to clean vs. non-clean × not controlled
- The **productivity of clean R&D**: innovations per dollar × not controlled

Illustration: two firms, identical total R&D

- Firm A: mostly non-clean R&D, high clean productivity → few clean patents
- Firm B: mostly clean R&D, low clean productivity → many clean patents

⇒ Firm B has more clean patents for a given amount of total R&D via a **composition effect**, not disclosure

→ **coefficient could be biased upward**

Comment #3: Asymmetry between input and output

- **What the IV resolves:** EPA enforcement leaves total R&D unchanged at $t = 0$ (Table B14)

  $\rightarrow$ instrumented variation in clean patents plausibly orthogonal to innovation investment

- **What the IV does not resolve:** total R&D unchanged $\not\Rightarrow$ **clean/non-clean split** unchanged.

  $\rightarrow$ If firms reallocate existing R&D toward green research under EPA pressure, the composition bias **persists in IV estimates**

Comment #4: IV strategy

Instrument: EPA enforcement actions against tech-close peers $j \rightarrow$ exogenous shock to disclosure incentives

- **Intuition:** regulatory pressure induces clean patent filings as compliance signals without increasing R&D

Exclusion restriction: EPA enforcement on j must affect i's value *only* through disclosed knowledge

- The paper documents that EPA-induced patents serve as signals to **regulators** (Table 8) and **investors** (Figure 3)
- If disclosure leads to sector-wide regulatory or investor reassessment, enforcement on j affects i through a **capital market / regulatory channel**, not through knowledge alone